GENERATIVE AI

Lab Experiment – 02:

1. **Develop a program to perform Bias Audit on pre-trained word embeddings to identify gender or racial stereotypes using the Word Embedding Association Test (WEAT).**
   Dataset: Common Crawl or Google News Embeddings.

Code:

```
!pip install gensim
import gensim.downloader as api
import numpy as np
from numpy.linalg import norm
model = api.load("glove-wiki-gigaword-100")
male_words = ["man", "male", "boy", "brother", "he", "son"]
female_words = ["woman", "female", "girl", "sister", "she", "daughter"]

career_words = ["career", "corporation", "salary", "office",
"professional"]
family_words = ["home", "parents", "children", "family", "marriage"]
def cosine_similarity(v1, v2):
    return np.dot(v1, v2) / (norm(v1) * norm(v2))
def association(word, A, B):

    w = model[word]

    sim_A = np.mean([cosine_similarity(w, model[a]) for a in A])
    sim_B = np.mean([cosine_similarity(w, model[b]) for b in B])

    return sim_A - sim_B
def weat_score(X, Y, A, B):

    score_X = np.sum([association(x, A, B) for x in X])
    score_Y = np.sum([association(y, A, B) for y in Y])

    return score_X - score_Y
score = weat_score(male_words, female_words, career_words,
family_words)

print("WEAT Bias Score:", score)
```

OUTPUT:

```
WEAT Bias Score: 0.41747725
```

```python
if score > 0:
    print("Embedding shows bias: Male associated with career.")
elif score < 0:
    print("Embedding shows bias: Female associated with career.")
else:
    print("No strong bias detected.")
```

OUTPUT:

```
Embedding shows bias: Male associated with career.
```